



2010
TRIAL
HIGHER SCHOOL CERTIFICATE

GIRRAWEE HIGH SCHOOL

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

Attempt Questions 1 – 10

All questions are of equal value

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Attempt Questions 1 – 10

All questions are of equal value

	Marks
Question 1 (12 marks).	
(a) Simplify $1 - \frac{p-q}{p+q}$.	2
(b) Solve $\frac{4x-5}{x} = 2$.	2
(c) Solve $ x-1 = 5$.	2
(d) Find the gradient of the tangent to the curve $y = x^3 - 4x$ at the point $(1, -3)$.	2
(e) Find the exact value of θ such that $2\sin\theta = 1$, where $0 \leq \theta \leq \frac{\pi}{2}$.	2
(f) Solve the equation $\ln x = 3$. Give your answer correct to three decimal places.	2

Marks

Question 2 (12 marks). Start on a SEPARATE page.

(a) Differentiate with respect to x :

(i) $x \tan x$. **2**

(ii) $(e^x + 1)^3$. **2**

(b) (i) Find $\int 4dx$. **1**

(ii) Find $\int \frac{2}{(x-5)^2} dx$. **2**

(iii) Evaluate $\int_0^3 \sqrt{5x+1} dx$. **3**

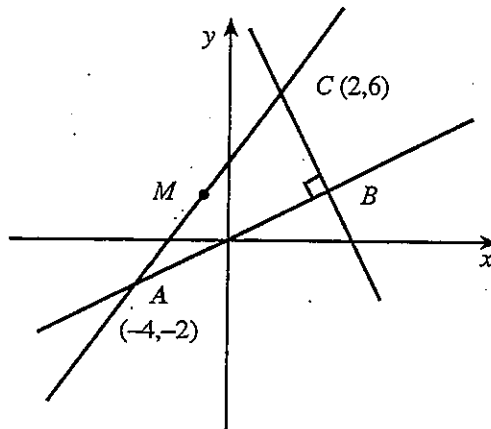
(c) Evaluate $\sum_{k=2}^5 \frac{(-1)^k}{k+1}$. **2**

Marks

Question 3 (12 marks). Start on a SEPARATE page.

- (a) An arithmetic series has 20 terms. The first term is 1 and the common difference is 7. Find the sum of the series. 2

- (b) NOT TO SCALE



- (i) Find the equation of the line AB, given that it passes through the origin. 2
- (ii) The line BC is perpendicular to AB. 2
- Show that its equation is $y = -2x + 10$.
- (iii) By solving the equations in (i) and (ii) above, find the coordinates of B. 2
- (iv) Find the length of AC. 1
- (v) Find the coordinates of M, the midpoint of AC. 1
- (vi) Explain why a circle, centre M, can be drawn to pass through A, B and C. 1
- (vii) Write down the equation of this circle. 1

Marks

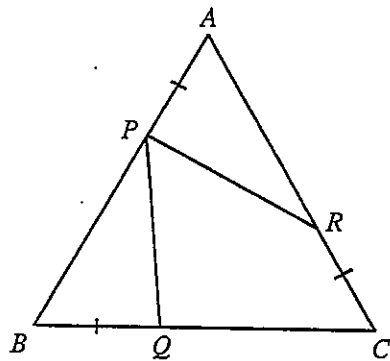
Question 4 (12 marks). Start on a SEPARATE page.

- (a) A man undertook to pay \$200 to a charity one year, \$150 the next year, three-quarters of \$150 the third year and so on until he died. What is the greatest sum of money the charity may expect from these donations ?

2

(b)

NOT TO SCALE



ΔABC is equilateral. $AP = BQ = CR$.

Copy or trace the diagram onto your answer page.

- (i) Prove that triangles APR and BQP are congruent .

4

- (ii) Prove that $\angle QPR = 60^\circ$.

3

- (iii) Prove that triangle PQR is equilateral .

3

Marks

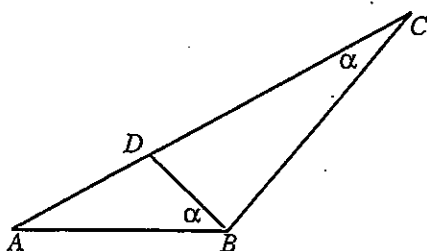
Question 5 (12 marks). Start on a SEPARATE page.

- (a) Find the values of k for which the quadratic equation $x^2 - k(x-1) + 3 = 0$ has equal roots .

3

(b)

NOT TO SCALE



Copy or trace the diagram onto your answer page.

- (i) Prove that triangles ABC and ADB are similar .
- (ii) If $AD = 4$ cm and $DC = 12$ cm, find the length of AB .
- (c) A certain soccer team has a probability of 0.5 of winning a match and a probability of 0.2 of drawing the match. If the team plays two matches, find the probability that it will :
- (i) draw both matches .
- (ii) win at least one match .
- (iii) not win either match .

2

2

1

2

2

Marks

Question 6 (12 marks). Start on a SEPARATE page.

- (a) An arc AB of a sector of a circle is of length $\frac{\pi}{4}$ metres and subtends an angle of 30° at the centre, O, of the circle.

(i) Find the length of the radius . 2

(ii) Find the area of the sector AOB . Give your answer correct to two decimal places . 1

(iii) Find the length of the chord AB. Give your answer correct to two decimal places. 2

(b) Find the perpendicular distance from the point $(2, -1)$ to the line $5x - 12y + 4 = 0$. 2

(c) Solve $2\log x = \log(5x + 6)$ 2

(d) The section of the curve $y = \sec x$, from $x = \frac{\pi}{4}$ to $x = \frac{\pi}{3}$ 3
is rotated about the x – axis. Find the exact value of the volume of the solid of revolution so formed.

Marks

Question 7 (12 marks). Start on a SEPARATE page.

- (a) Solve $-x^2 + 13x - 36 = 0$. 2
- (b) Find the equation of the tangent to the parabola $y = -x^2 + 13x - 36$ at the point where $x = 6$. 3
- (c) Draw a diagram showing the parabola and the tangent. Shade the region bounded by the parabola, the tangent and the x - axis . 3
- (d) Find the area shaded in the diagram above. 4

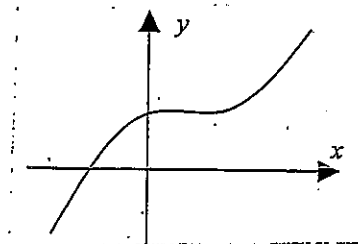
Question 8 (12 marks). Start on a SEPARATE page.

- (a) Let $f(x) = \frac{1}{3}x^3 + x^2 - 3x + 5$.
- (i) Find the stationary points and determine their nature. 4
- (ii) Find any points of inflection. 2
- (iii) Sketch the graph of $f(x)$. 1
- (iv) For what values of x is $f(x)$ concave upwards ? 1
- (b) The mass, M , in grams of a radioactive substance is expressed as $M = 175e^{-kt}$ where k is a positive constant and t the time in days. The mass of the substance halved in 6 days.
- (i) Find the value of k correct to 5 decimal places. 2
- (ii) At what rate is the mass disintegrating after 10 days ? 2

Marks

Question 9 (12 marks). Start on a SEPARATE page.

- (a) The diagram shows the graph of a function $y = f(x)$.



Sketch the graph of $y = f'(x)$.

2

- (b) The gradient function of a curve is given by $6x - \frac{2}{2x-1}$.

2

Find the equation of the curve if it passes through the point $(1, 7)$.

- (c) An amount of \$10 000 is borrowed and an interest rate of 1% per month is charged monthly. An amount M is repaid every month.

- (i) If A_n is the amount owing after n months, show that

4

$$A_n = \$10000(1.01)^n - M \left(\frac{1.01^n - 1}{0.01} \right).$$

- (ii) Find the value of M , to the nearest cent, if the loan is repaid at the end of 5 years.

2

- (iii) How much extra, in total, will be repaid if the loan is taken over 7 years?

2

Marks

Question 10 (12 marks). Start on a SEPARATE page.

- (a) Use Simpson's rule, with five function values, to approximate

3

$$\int_0^2 \sqrt{x^2 + 4} dx.$$

- (b) A particle, initially at the origin, moves so that after
- t
- seconds its

$$\text{velocity, } v \text{ m/s, is given by } v = \frac{6}{\sqrt{2t+1}}.$$

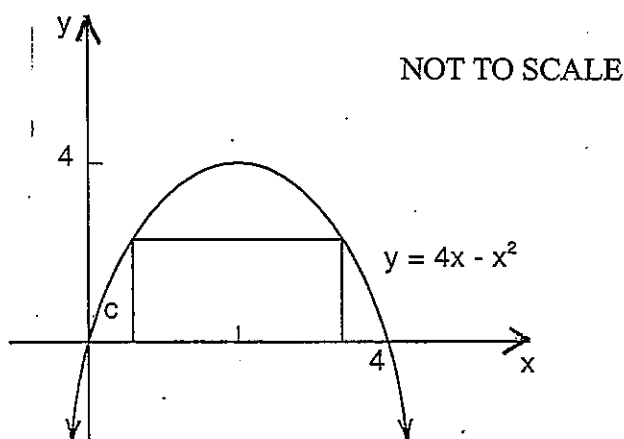
- (i) Show that the position of the particle is given by
- $x = 6\sqrt{2t+1} - 6$
- .

1

- (ii) Find the particle's average velocity in moving from
- $x = 0$
- to
- $x = 24$
- .

2

- (c) A rectangle has two of its vertices on the curve
- $y = 4x - x^2$
- and the other two vertices on the
- x
- axis in the interval
- $0 \leq x \leq 4$
- as shown in the diagram below.



- (i) If the height of the rectangle is
- c
- cm, show that its area is
- $2c\sqrt{4-c}$
- square centimetres.

3

- (ii) Show that the greatest value of this area is

3

$$\frac{32\sqrt{3}}{9} \text{ square centimetres.}$$

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Mathematics Trial HSC 2010 Solutions.

Question 1

$$\begin{aligned} (a) \quad 1 - \frac{p-q}{p+q} \\ = \frac{p+q - p+q}{p+q} \\ = \frac{2q}{p+q} \# \quad (2) \end{aligned}$$

$$\begin{aligned} (b) \quad \frac{4x-5}{x} &= 2 \\ 4x-5 &= 2x \\ 4x-2x &= 5 \\ 2x &= 5 \\ \therefore x &= \frac{5}{2} \text{ or } 2\frac{1}{2} \# \quad (2) \end{aligned}$$

$$\begin{aligned} (c) \quad |x-1| &= 5 \\ x-1 &= 5 \text{ or } -(x-1) = 5 \\ x &= 6 \text{ or } -x+1 = 5 \\ \therefore x &= 6 \text{ or } x = -4 \# \quad (2) \end{aligned}$$

$$\begin{aligned} (d) \quad y &= x^3 - 4x \\ \therefore \frac{dy}{dx} &= 3x^2 - 4 \\ \text{when } x=1, \frac{dy}{dx} &= 3 \times 1^2 - 4 \\ \therefore \frac{dy}{dx} &= -1 \# \quad (2) \end{aligned}$$

$$\begin{aligned} (e) \quad 2 \sin \theta &= 1 \\ \therefore \sin \theta &= \frac{1}{2} \\ \therefore \theta &= \frac{\pi}{6} \# \quad (2) \end{aligned}$$

$$\begin{aligned} (f) \quad \ln x &= 3 \\ \therefore x &= e^3 \\ \therefore x &= 20.086 \# \quad (2) \end{aligned}$$

Question 2

$$\begin{aligned} (a) \quad (i) \quad \frac{d}{dx}(x \tan x) \\ = x \times \frac{d}{dx}(\tan x) + \tan x \times \frac{d}{dx}(x) \\ = x \sec^2 x + \tan x \# \quad (2) \end{aligned}$$

$$\begin{aligned} (ii) \quad \frac{d}{dx}[(e^x+1)^3] \\ = 3(e^x+1)^2 \times \frac{d}{dx}(e^x+1) \\ = 3e^x(e^x+1)^2 \# \quad (2) \end{aligned}$$

$$(b) \quad (i) \quad \int 4 dx = 4x + C \# \quad (1)$$

$$\begin{aligned} (ii) \quad \int \frac{2}{(x-5)^2} dx \\ = 2 \int (x-5)^{-2} dx \\ = -2(x-5)^{-1} + C \\ = \frac{-2}{x-5} + C \# \quad (2) \end{aligned}$$

$$\begin{aligned} (iii) \quad \int_0^3 \sqrt{5x+1} dx \\ = \left[\frac{(5x+1)^{3/2}}{5 \times 3/2} \right]_0^3 \\ = \left[\frac{2}{15} (5x+1)^{3/2} \right]_0^3 \\ = \frac{2}{15} \times 16^{3/2} - \frac{2}{15} \times 1^{3/2} \\ = \frac{2}{15} \times 64 - \frac{2}{15} \\ = 8\frac{2}{5} \# \quad (3) \end{aligned}$$

$$(c) \sum_{k=2}^5 \frac{(-1)^k}{k+1}$$

$$= \frac{(-1)^2}{2+1} + \frac{(-1)^3}{3+1} + \frac{(-1)^4}{4+1} + \frac{(-1)^5}{5+1}$$

$$= \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6}$$

$$= \frac{7}{60} \# \quad (2)$$

Question 3

(a) $n=20, a=1, d=7.$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\therefore S_{20} = \frac{20}{2} [2 \times 1 + (20-1) \times 7]$$

$$= 10 (2 + 133)$$

$$= 1350 \# \quad (2)$$

(b) (i) $A(-4, -2); O(0, 0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 - (-2)}{0 - (-4)}$$

$$= \frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{1}{2}(x - 0)$$

$$y = \frac{1}{2}x \# \quad (2)$$

(ii) $m_{AB} = \frac{1}{2}$

$$\therefore \frac{1}{2} \times m_{BC} = -1$$

$$\therefore m_{BC} = -2$$

equation of BC is

$$y - 6 = -2(x - 2)$$

$$y - 6 = -2x + 4$$

$$\therefore y = -2x + 10 \# \quad (2)$$

(iii) $y = \frac{1}{2}x$

$$\therefore x = 2y$$

Sub. $x = 2y$ into $y = -2x + 10$

$$\therefore y = -2(2y) + 10$$

$$y = -4y + 10$$

$$5y = 10$$

$$\therefore y = 2$$

$$\therefore x = 2 \times 2 = 4$$

$$\therefore B(4, 2) \# \quad (2)$$

(iv) $AC = \sqrt{(2+4)^2 + (6+2)^2}$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$\therefore AC = 10 \text{ units} \# \quad (1)$$

(v) $M\left(\frac{2-4}{2}, \frac{6-2}{2}\right)$

$$= M(-1, 2) \# \quad (1)$$

(vi) $MB = 4 + 1 = 5 \text{ units}$

$$MA = MC = 5 \text{ units} = MB$$

$\therefore$ a circle, centre M, passes through A, B and C. $\# \quad (1)$

(vii) $(x+1)^2 + (y-2)^2 = 25 \# \quad (1)$

Question 4

(a) $\$200 + \$150 + \$\frac{3}{4} \times 150 + \dots$

a geometric series with

$$a = \$200, r = \frac{3}{4}$$

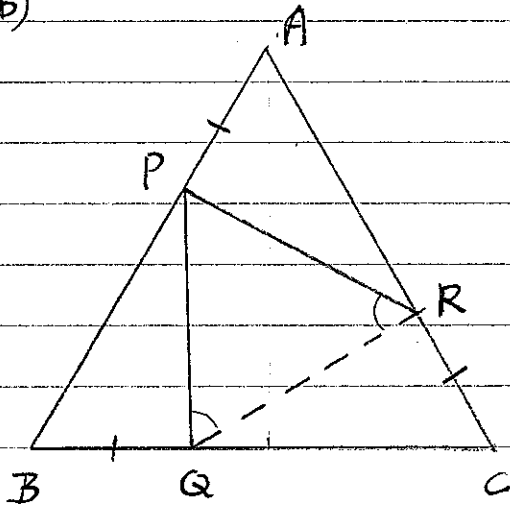
Since $|r| < 1$, then

$$S = \frac{a}{1-r}$$

$$= \frac{\$200}{1-\frac{3}{4}}$$

$$= \$800 \quad \# \quad (2)$$

(b)



(i) In Δ s APR and BQP.

$$AC = AB \text{ (given)}$$

$$RC = PA \text{ (given)}$$

$$AC - RC = AB - PA$$

$$\therefore AR = BP$$

$$AP = BQ \text{ (given)}$$

$$\angle PAR = \angle QBR \text{ (} 60^\circ \text{, equilateral } \Delta \text{)}$$

$$\therefore \Delta APR \equiv \Delta BQP \text{ (SAS)} \quad \#$$

(4)

(ii)

$$\angle ARP = \angle BPQ \text{ (matching } \angle \text{s of congruent } \Delta \text{s)}$$

$$\angle APR + \angle ARP = 120^\circ \text{ (} \angle \text{ sum of } \Delta APR \text{)}$$

$$\therefore \angle APR + \angle BPQ + \angle QPR = 180^\circ \text{ (straight } \angle \text{)}$$

$$\therefore 120^\circ + \angle QPR = 180^\circ$$

$$\therefore \angle QPR = 60^\circ \quad \# \quad (3)$$

(iii) Join QR

$$PQ = PR \text{ (matching sides of congruent } \Delta \text{s)}$$

$$\therefore \angle PQR = \angle PRQ \text{ (} \angle \text{s opposite equal sides of isosceles } \Delta PQR \text{)}$$

$$\text{But } \angle PQR + \angle PRQ = 120^\circ \text{ (} \angle \text{ sum of } \Delta PQR \text{)}$$

$$\therefore \angle PQR = \angle PRQ = \angle QPR$$

$$\therefore \Delta PQR \text{ is equilateral.} \quad \#$$

(3)

Question 5

$$(a) \quad x^2 - k(x-1) + 3 = 0$$

$$x^2 - kx + k + 3 = 0.$$

$$\Delta = b^2 - 4ac$$

$$= (-k)^2 - 4(k+3)$$

$$= k^2 - 4k - 12$$

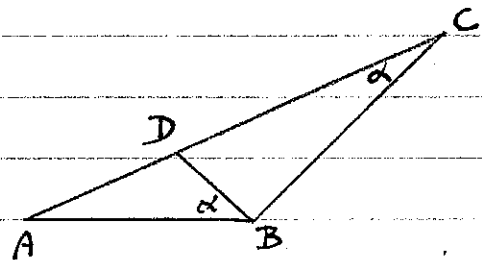
$$\text{For equal roots, } \Delta = 0$$

$$\therefore k^2 - 4k - 12 = 0$$

$$(k-6)(k+2) = 0$$

$$\therefore k = 6 \text{ or } k = -2. \quad \# \quad (3)$$

(b)



(i) In Δ s ABC and ADB.

$$\angle ACB = \angle ABD \text{ (given)}$$

$$\angle BAC \text{ is common}$$

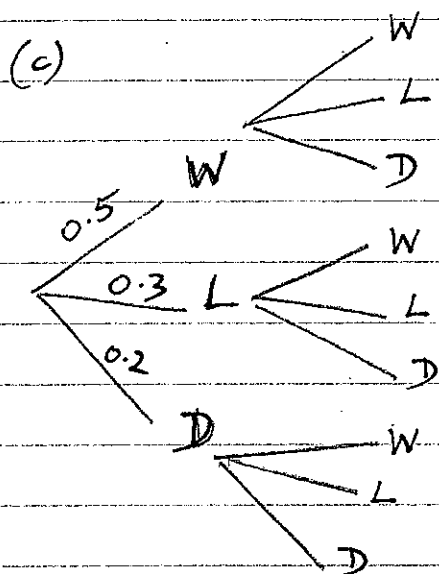
$$\therefore \triangle ABC \parallel \triangle ADB \text{ (equiangular)} \\ \# \text{ (2)}$$

$$(ii) \frac{AB}{AC} = \frac{AD}{AB} \text{ (matching sides of similar } \triangle s \text{ are proportional)}$$

$$\therefore \frac{AB}{16} = \frac{4}{AB}$$

$$AB^2 = 64$$

$$\therefore AB = 8 \text{ cm. } \# \text{ (2)}$$



$$(i) P(DD) = 0.2 \times 0.2 \\ = 0.04 \# \text{ (1)}$$

$$(ii) P(\text{win at least 1 match}) \\ = 1 - P(LL + LD + DL + DD) \\ = 1 - [0.3^2 + 2 \times 0.3 \times 0.2 + 0.2^2] \\ = 1 - 0.25 \\ = 0.75 \# \text{ (2)}$$

$$(iii) P(\text{not win either match}) \\ = P(LL + LD + DL + DD)$$

$$= 0.25 \# \text{ (2)}$$

Question 6

$$(a) (i) l = r\theta$$

$$\therefore \frac{\pi}{4} = r \times \frac{\pi}{6}$$

$$\therefore r = \frac{\pi}{4} \times \frac{6}{\pi}$$

$$\therefore r = 1.5 \text{ m. } \# \text{ (2)}$$

$$(ii) A = \frac{1}{2} r^2 \theta \\ = \frac{1}{2} \times 1.5^2 \times \frac{\pi}{6} \\ = 0.59 \text{ m}^2 \# \text{ (1)}$$

$$(iii) AB^2 = 1.5^2 + 1.5^2 - 2 \times 1.5^2 \times \cos 30^\circ \\ = 4.5 - 4.5 \cos 30^\circ \\ = 0.602885083... \\ \therefore AB = 0.78 \text{ m } \# \text{ (2)}$$

$$(b) d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ = \frac{|5 \times 2 - 12 \times -1 + 4|}{\sqrt{5^2 + (-12)^2}} \\ = \frac{26}{13} \\ = 2 \text{ units } \# \text{ (2)}$$

$$(c) 2 \log x = \log(5x+6) \\ \log x^2 = \log(5x+6) \\ \therefore x^2 = 5x+6 \\ x^2 - 5x - 6 = 0 \\ (x-6)(x+1) = 0 \\ \therefore x = 6 \text{ or } x = -1$$

$$\therefore x=6 \text{ since } x>0 \text{ \# (2)}$$

$$\begin{aligned} \text{(d)} \quad V &= \pi \int_{\pi/4}^{\pi/3} y^2 dx \\ &= \pi \int_{\pi/4}^{\pi/3} \sec^2 x dx \\ &= \pi [\tan x]_{\pi/4}^{\pi/3} \end{aligned}$$

$$= \pi \left(\tan \frac{\pi}{3} - \tan \frac{\pi}{4} \right)$$

$$= \pi (\sqrt{3} - 1) \text{ units}^3 \text{ \# (3)}$$

Question 7

$$\text{(a)} \quad -x^2 + 13x - 36 = 0$$

$$\therefore x^2 - 13x + 36 = 0$$

$$(x-4)(x-9) = 0$$

$$\therefore x=4 \text{ or } x=9 \text{ \# (2)}$$

$$\text{(b)} \quad y = -x^2 + 13x - 36$$

$$\text{at } x=6; y = -6^2 + 13 \times 6 - 36 = 6$$

$$\frac{dy}{dx} = -2x + 13$$

$$\text{at } x=6, \frac{dy}{dx} = -2 \times 6 + 13 = 1.$$

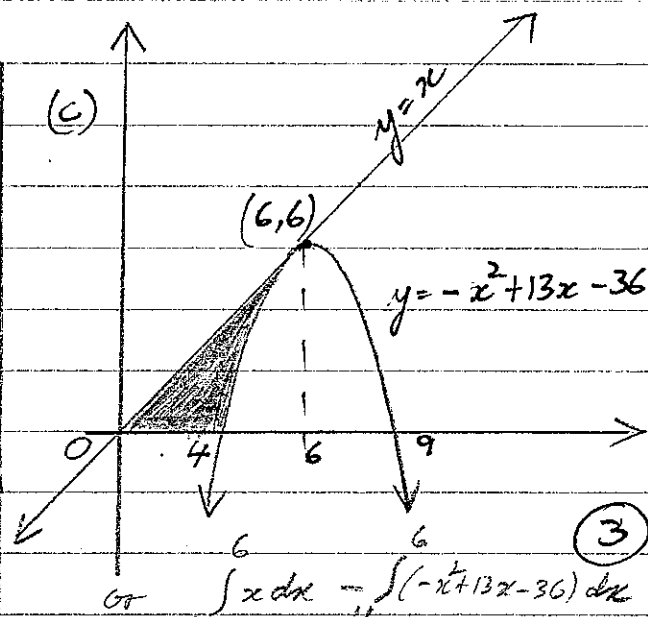
equation of tangent is

$$y - 6 = 1(x - 6)$$

$$y - 6 = x - 6$$

$$\therefore y = x \text{ \# (3)}$$

(c)



(d) Area shaded;

$$= \frac{1}{2} \times 6 \times 6 - \int_4^6 (-x^2 + 13x - 36) dx$$

$$= 18 - \left[-\frac{x^3}{3} + \frac{13x^2}{2} - 36x \right]_4^6$$

$$= 18 - \left[-\frac{6^3}{3} + \frac{13 \times 6^2}{2} - 36 \times 6 \right]$$

$$= \left[-\frac{4^3}{3} + \frac{13 \times 4^2}{2} - 36 \times 4 \right]$$

$$= 18 - [-54] - [61 \frac{2}{3}]$$

$$= 18 + 54 - 61 \frac{2}{3}$$

$$= 10 \frac{2}{3} \text{ units}^2 \text{ \# (4)}$$

Question 8

$$\text{(a) (i)} \quad f(x) = \frac{1}{3}x^3 + x^2 - 3x + 5$$

$$f'(x) = x^2 + 2x - 3$$

$$f''(x) = 2x + 2$$

$f'(x) = 0$ for stationary points

$$\therefore x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$\therefore x = -3 \text{ or } x = 1.$$

When $x = -3$,

$$\begin{aligned} f(x) &= \frac{1}{3}(-3)^3 + (-3)^2 - 3 \times -3 + 5 \\ &= -9 + 9 + 9 + 5 \\ &= 14. \end{aligned}$$

when $x = -3$,

$$f''(x) = 2x - 3 + 2 = -4.$$

$\therefore (-3, 14)$ is a maximum turning point. #

When $x = 1$,

$$\begin{aligned} f(x) &= \frac{1}{3} + 1 - 3 + 5 \\ &= 3\frac{1}{3}. \end{aligned}$$

when $x = 1$,

$$f''(x) = 2 \times 1 + 2 = 4.$$

$\therefore (1, 3\frac{1}{3})$ is a minimum turning point. # ④

(ii) For points of inflection
 $f''(x) = 0$

$$\begin{aligned} \therefore 2x + 2 &= 0 \\ 2x &= -2 \\ \therefore x &= -1 \end{aligned}$$

when $x = -1$,

$$\begin{aligned} f(x) &= \frac{1}{3}(-1)^3 + (-1)^2 - 3(-1) + 5 \\ &= -\frac{1}{3} + 1 + 3 + 5 \\ &= 8\frac{2}{3}. \end{aligned}$$

$\therefore (-1, 8\frac{2}{3})$ is a possible point of inflection.

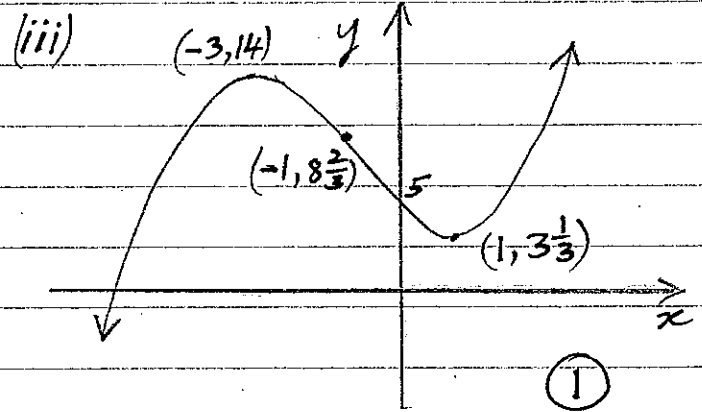
Test for concavity:

$$\begin{aligned} \text{when } x = -0.9, f''(x) &= 2x - 0.9 + 2 \\ &= 0.2 \end{aligned}$$

$$\begin{aligned} \text{when } x = -1.1, f''(x) &= 2x - 1.1 + 2 \\ &= -0.2 \end{aligned}$$

Since concavity changes about $x = -1$,

$(-1, 8\frac{2}{3})$ is a point of inflection. # ②



(iv) Concave upwards when
 $f''(x) > 0$.

$$\therefore 2x + 2 > 0$$

$$2x > -2$$

$$\therefore x > -1. \# \text{ ①}$$

$$\begin{aligned} \text{(b) (i) } M &= 175e^{-kt} \\ \therefore 87.5 &= 175e^{-6k} \\ 0.5 &= e^{-6k} \end{aligned}$$

$$\ln 0.5 = -6k \ln e$$

$$\therefore k = \frac{\ln 0.5}{-6} \quad (\ln e = 1)$$

$$\therefore k = 0.11552 \# \text{ ②}$$

$$\text{(ii) } \frac{dM}{dt} = -175ke^{-kt}$$

$\therefore$ when $t = 10$,

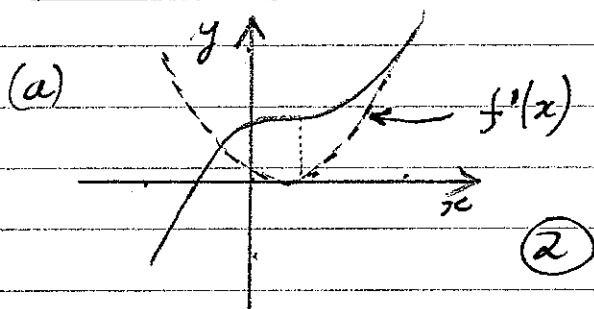
$$\frac{dM}{dt} = -175 \times \frac{\ln 0.5}{-6} e^{-10 \frac{\ln 0.5}{-6}}$$

$$= -6.367890692 \dots$$

$$= -6.4 \text{ g/day}$$

i.e. disintegrating at 6.4 g/day. # ②

Question 9



(b) $\frac{dy}{dx} = 6x - \frac{2}{2x-1}$

$\therefore y = 3x^2 - \ln(2x-1) + C$

Sub. (1,7)

$\therefore 7 = 3 - \ln 1 + C$

$\therefore C = 4$

$\therefore y = 3x^2 - \ln(2x-1) + 4$ # (2)

(c)

(i) \$10000, $t = 1.01$

Amount owing after 1 month,

A_1 , is given by

$A_1 = \$10000 \times 1.01 - M$

Amount owing after 2 months,

A_2 , is given by

$A_2 = A_1(1.01) - M$

$= [\$10000(1.01) - M] 1.01 - M$

$= \$10000(1.01)^2 - M(1 + 1.01)$

Amount owing after 3 months,

A_3 , is given by

$A_3 = A_2(1.01) - M$

$= [\$10000(1.01)^2 - M(1 + 1.01)]$

$\times 1.01 - M$

$= \$10000(1.01)^3 - M(1 + 1.01 + 1.01^2)$

$\therefore$

$A_n = \$10000(1.01)^n - M(1 + 1.01 + \dots + 1.01^{n-1})$

Now $1 + 1.01 + 1.01^2 + \dots + 1.01^{n-1}$

is a geometric series with
 $a=1$, $r=1.01$, n terms

Now $S = a \frac{(r^n - 1)}{r - 1}$

$\therefore S = 1 \frac{(1.01^n - 1)}{1.01 - 1}$

$\therefore A_n = \$10000(1.01)^n - M \frac{(1.01^n - 1)}{0.01}$

(4)

(ii) $A_{60} = 0$

$\therefore \$10000(1.01)^{60} = M \frac{(1.01^{60} - 1)}{0.01}$

$\therefore M = \$ \frac{10000(1.01)^{60} \times 0.01}{1.01^{60} - 1}$

$= \$222.44$ # (2)

(iii) $M = \$ \frac{10000(1.01)^{84} \times 0.01}{1.01^{84} - 1}$

$= \$176.53$

$\therefore$ repayments over 7 years

$= \$176.53 \times 84$

$= \$14828.52$

Repayments over 5 years

$= \$222.44 \times 60$

$= \$13346.40$

$\therefore$ extra $= \$14828.52 - \13346.40

$= \$1482.12$ # (2)

Question 10

(a)

x	0	0.5	1	1.5	2
$f(x)$	2	$\sqrt{4-25}$	$\sqrt{5}$	2.5	$\sqrt{8}$
w	1	4	2	4	1

$$A \div (2-0) \times \left\{ \frac{2+4\sqrt{4-25}+2\sqrt{5}+10+\sqrt{8}}{1+4+2+4+1} \right\}$$

$$= 4.591129055 \dots$$

$$= 4.6 \text{ \# } (3)$$

(b) (i) $v = \frac{6}{\sqrt{2t+1}} = 6(2t+1)^{-1/2}$

$$\begin{aligned} \therefore x &= \int 6(2t+1)^{-1/2} dt \\ &= \frac{6(2t+1)^{1/2}}{\frac{1}{2} \times 2} + c \\ &= 6\sqrt{2t+1} + c \end{aligned}$$

when $t=0$, $x=0$

$$\therefore 0 = 6 + c$$

$$\therefore c = -6$$

$$\therefore x = 6\sqrt{2t+1} - 6 \text{ \# } (1)$$

(ii) When $x=0$, $t=0$

when $x=24$,

$$24 = 6\sqrt{2t+1} - 6$$

$$\therefore 30 = 6\sqrt{2t+1}$$

$$\therefore 5 = \sqrt{2t+1}$$

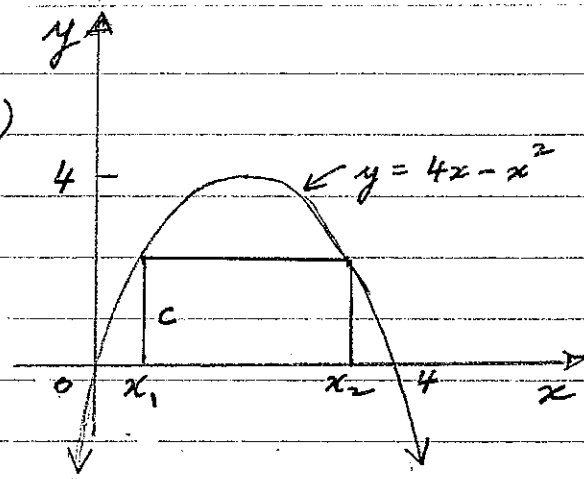
$$\therefore 25 = 2t+1$$

$$2t = 24$$

$$\therefore t = 12$$

$$\begin{aligned} \therefore \text{average velocity} &= \frac{24-0}{12-0} \\ &= 2 \text{ m/s \# } (2) \end{aligned}$$

(c)



(i) If $y=c$, then $c = 4x - x^2$
 $\therefore x^2 - 4x + c = 0$

$$\therefore x = \frac{4 \pm \sqrt{16-4c}}{2}$$

$$= \frac{4 \pm 2\sqrt{4-c}}{2}$$

$$= 2 \pm \sqrt{4-c}$$

$$\therefore x = 2 + \sqrt{4-c} \text{ or } 2 - \sqrt{4-c}$$

i.e. $x_1 = 2 - \sqrt{4-c}$ and

$$x_2 = 2 + \sqrt{4-c}$$

$\therefore$ Length of rectangle

$$= (2 + \sqrt{4-c}) - (2 - \sqrt{4-c}) \text{ cm}$$

$$= 2\sqrt{4-c} \text{ cm}$$

$\therefore$ Area of rectangle

$$= 2c\sqrt{4-c} \text{ cm}^2 \text{ \# } (3)$$

(ii) Let A represent the area (in cm^2).

$$\therefore A = 2c\sqrt{4-c}$$

$$\therefore \frac{dA}{dc} = 2c \times \frac{1}{2}(4-c)^{-1/2} \times -1 + (4-c)^{1/2} \times 2$$

$$= \frac{-c}{\sqrt{4-c}} + 2\sqrt{4-c}$$

For min./max A , $\frac{dA}{dc} = 0$

$$\therefore 2\sqrt{4-c} - \frac{c}{\sqrt{4-c}} = 0$$

$$\therefore 2\sqrt{4-c} = \frac{c}{\sqrt{4-c}}$$

Cross-multiplying,

$$2(4-c) = c$$

$$8 - 2c = c$$

$$8 = 3c$$

$$\therefore c = \frac{8}{3} = 2\frac{2}{3}$$

c	2	$2\frac{2}{3}$	3
$\frac{dA}{dc}$	$\sqrt{2}$	0	-1

Check: when $c = 2$

$$\begin{aligned} \frac{dA}{dc} &= 2\sqrt{4-2} - \frac{2}{\sqrt{4-2}} \\ &= 2\sqrt{2} - \frac{2 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} \\ &= 2\sqrt{2} - \sqrt{2} \\ &= \sqrt{2} > 0. \end{aligned}$$

when $c = 3$,

$$\begin{aligned} \frac{dA}{dc} &= 2\sqrt{4-3} - \frac{3}{\sqrt{4-3}} \\ &= 2 - 3 \\ &= -1 < 0. \end{aligned}$$

$\therefore$ maximum A occurs

when $c = 2\frac{2}{3}$ cm

$$\therefore A = 2 \times \frac{8}{3} \sqrt{4 - \frac{8}{3}}$$

$$= \frac{16}{3} \sqrt{\frac{4}{3}}$$

$$= \frac{16}{3} \times \frac{2}{\sqrt{3}}$$

$$= \frac{32}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{32\sqrt{3}}{9} \text{ cm}^2 \quad \# \quad (3)$$